

BEVFormer Training Pipeline and Code-Level Implementation Analysis

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Abstract

We first summarize the overall detector architecture and clarify how BEVFormer extends multi-view 3D detectors with a spatiotemporal transformer operating on a bird's-eye-view (BEV) grid. We then describe how a temporal queue of multi-camera images is used to construct history BEV features, how current-frame image features are extracted, and how these are fused with ego-motion information from the CAN bus. Next, we explain how the BEVFormer head instantiates BEV queries and object queries, how the BEV-aware transformer encoder-decoder processes them, and how the head converts the decoder outputs into class scores and 3D bounding box predictions. Finally, we discuss the DETR-style loss computation and provide qualitative observations about training behavior, ablations on history BEV, and practical implications for debugging and deployment.

The resulting analysis serves as a set of structured study notes that make the BEVFormer training code easier to understand, modify, and extend, especially in the context of edge deployment and NPU-friendly design. *This report presents a detailed, code-level analysis of the BEVFormer training pipeline as implemented in a PyTorch/MMDetection3D-based codebase. Instead of focusing only on high-level diagrams, we trace how a single training mini-batch flows through concrete functions such as `forward_train`, `obtain_history_bev`, `extract_feat`, `BEVFormerHead.forward`, and `PerceptionTransformer.get_bev_features`.*

We first summarize the overall detector architecture and clarify how BEVFormer extends multi-view 3D detectors with a spatiotemporal transformer operating on a bird's-eye-view (BEV) grid. We then describe how a temporal queue of multi-camera images is used to construct history BEV features, how current-frame image features are extracted, and how these are fused with ego-motion information from the CAN bus. Next, we explain how the BEVFormer head instantiates BEV queries and object queries, how the BEV-aware transformer encoder-decoder processes them, and how the head converts the decoder outputs into class scores and 3D bounding box predictions. Finally, we discuss the DETR-style loss computation and provide qualitative observations about training behavior, ablations on history BEV, and practical implications for debugging and deployment.

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1. Introduction

Camera-only 3D perception has become a key component in modern autonomous driving systems. Among various representations, the bird's-eye-view (BEV) has emerged as a particularly convenient

space for unifying multi-camera imagery, performing object detection, and integrating additional scene understanding tasks such as map segmentation and lane detection. BEVFormer is a transformer-based architecture that directly learns a BEV representation from synchronized multi-camera images using spatiotemporal self-attention.

While the original BEVFormer paper focuses on the conceptual architecture and experimental results, implementing the model in a production-ready codebase introduces additional complexity. The training pipeline must handle temporal frame queues, multi-level features from backbone and FPN, ego-motion information from CAN bus signals, and DETR-style loss computation. For practitioners who wish to debug, modify, or deploy BEVFormer, it is therefore essential to understand not only the high-level idea but also the detailed execution flow of the training code.

The main goal of this report is to provide such a code-level explanation. We examine the training path implemented in three main modules:

- the detector class `BEVFormer`,
- the detection head `BEVFormerHead`,
- and the BEV-aware transformer `PerceptionTransformer`.

We reconstruct the training logic as a sequence of conceptual blocks and tensor transformations, closely tied to specific function names and arguments. The report is written as if it were a set of study notes prepared after manually reading through the codebase.

The remainder of this document is organized as follows. Section 2 describes the methods, including the structure of the codebase and the internal logic of key functions. Section 3 summarizes several empirical observations obtained from running BEVFormer training with and without temporal BEV history. Section 4 provides a discussion of design choices and potential extensions. Section 5 concludes the report.

2. Methods

2.1 Codebase Overview

The implementation analyzed in this report follows the `MMDetection3D` design pattern and is split across several files:

- **Detector:** `BEVFormer` (e.g., in `bevformer.py`). This class derives from a multi-view detector base class and implements the high-level training and inference interfaces. It orchestrates:
 - handling of image inputs and temporal queues,
 - construction of history BEV features using previous frames,
 - extraction of current-frame image features with backbone and neck,
 - calling the detection head for training (`forward_pts_train`).
- **Detection head:** `BEVFormerHead` (e.g., in `bevformer_head.py`). This head extends a DETR-style head and:
 - defines BEV queries and object queries as learnable embeddings,
 - interfaces with the BEV-aware transformer encoder–decoder,
 - interprets decoder outputs as 3D bounding boxes and class scores,

- computes losses using DETR-like Hungarian matching.
- **BEV-aware transformer:** *PerceptionTransformer* (e.g., in `transformer.py`). This module:
 - fuses multi-level image features from all cameras into a BEV representation,
 - integrates temporal information by aligning history BEV with current frames using CAN bus signals,
 - runs a decoder where object queries attend to BEV features to produce per-query representations.

In the following subsections, we explain how these modules interact along the training path. We deliberately keep the description as close as possible to the actual function names and arguments.

2.2 Training Entry Point: `forward_train`

During training, the *MMDetection3D* runner calls the detector's `forward_train` method. Its signature contains many arguments; here we highlight the most important ones:

```
forward_train(img, img metas, gt_bboxes_3d, gt_labels_3d, ...)
```

where:

- *img* is a 6D tensor of shape (B, T, N_{cam}, C, H, W) , representing a temporal queue of T frames for each of N_{cam} cameras in a batch of size B .
- *img_metas* is a list of lists of dictionaries. For each batch element, *img_metas*[*b*][*t*] stores metadata for frame *t*, including intrinsic and extrinsic calibration, image shape, CAN bus signals, and scene tokens.
- *gt_bboxes_3d* and *gt_labels_3d* are lists of 3D bounding boxes and labels for the current frame.

The logic of `forward_train` can be summarized as follows:

- (a) Compute the queue length as `len_queue = img.size(1)`.
- (b) Split the image queue into:

```
prev_img = img[:, :-1, ...],
img_cur = img[:, -1, ...].
```

Thus, previous frames *prev_img* have shape $(B, T - 1, N_{cam}, C, H, W)$, and the current frame *img_cur* has shape (B, N_{cam}, C, H, W) .

- (c) Deep-copy the full metadata list to *img_metas_queue* and pass it with *prev_img* to `obtain_history_be` in order to construct a history BEV feature *prev_bev*.
- (d) Extract the current frame's metadata as *img_metas_cur* = [each[*len_queue* - 1] for each in *img_metas*]. If the metadata indicates that no previous BEV exists (for example, at the beginning of a new scene), *prev_bev* is reset to `None`.

- (e) Call `extract_feat` on `img_cur` to obtain multi-level image features for the current frame.
- (f) Call `forward_pts_train` with the extracted features, ground truths, current metadata, and `prev_bev`. This function delegates to the BEVFormer head and returns a dictionary of loss terms.
- (g) Return the loss dictionary to the training loop for backpropagation.

In this way, `forward_train` wraps all training-related operations: it prepares temporal context, extracts features, and invokes the head for loss computation.

2.3 History BEV Construction: `obtain_history_bev`

2.3.1 Purpose

The function `obtain_history_bev` accumulates information from previous frames into a single BEV feature tensor. This tensor, denoted as `prev_bev`, acts as a memory of the past and allows BEVFormer to perform spatiotemporal reasoning during training.

2.3.2 Procedure

Given a queue of images `imgs_queue` and a corresponding metadata list `img metas_list`, the procedure is:

1. Temporarily switch the detector to evaluation mode using `self.eval()` and wrap the computation in `torch.no_grad()`. This ensures that the history path does not contribute to gradient computation and saves GPU memory.
2. Flatten the temporal dimension into the batch dimension and call `extract_feat` with the argument `len_queue = T`. This produces a list of multi-level features $\{\text{mlvl_feats}[l]\}$, where each tensor has shape $(B, T, N_{\text{cam}}, C_l, H_l, W_l)$.
3. Initialize `prev_bev = None`. Then iterate over the temporal index $i = 0, \dots, T - 1$. For frame i :

- Construct `img_metas` as `[each[i] for each in img_metas_list]`.
- If `img_metas[0]["prev_bev_exists"]` is `False`, reset `prev_bev = None`. This handles scene boundaries.
- For each feature level l , extract the frame slice `mlvl_feats[l][:, i]`, which has shape $(B, N_{\text{cam}}, C_l, H_l, W_l)$. Collect them into a list `img_feats`.
- Call the BEVFormer head in encoder-only mode:

`prev_bev ← pts_bbox_head.forward(img_feats, img_metas, prev_bev, only_bev=True)`

Internally, this runs the BEV encoder part of the PerceptionTransformer, fusing the new frame into the existing BEV memory.

4. After all frames are processed, restore training mode with `self.train()` and return the final `prev_bev`.

This design has two advantages. First, by using `no_grad()` and evaluation mode, the history computation does not slow down or destabilize training. Second, by reusing the same encoder logic as the main frame, the model learns a unified way to integrate past and present information.

2.4 Image Feature Extraction: `extract_img_feat` and `extract_feat`

The detector exposes a helper method `extract_feat`, which primarily calls `extract_img_feat`. The latter handles backbone and neck processing while keeping camera and batch dimensions explicit.

For a current frame tensor $\text{img_cur} \in \mathbb{R}^{B \times N_{\text{cam}} \times C \times H \times W}$, the steps are:

1. Reshape to $(B \cdot N_{\text{cam}}, C, H, W)$ to form a standard image batch.
2. Optionally apply grid mask augmentation to each image.
3. Pass the batch through an image backbone (for example, ResNet) to obtain a list of feature maps at different spatial resolutions.
4. If a feature pyramid neck (FPN) is present, feed the backbone outputs into the neck to obtain multi-level features with unified channel dimensions.
5. Reshape each level back into the form $(B, N_{\text{cam}}, C_l, H_l, W_l)$ for subsequent processing by the BEV transformer.

The output of `extract_feat` is therefore a list $\{\text{mlvl_feats}[l]\}$ of geospatially organized image features ready to be unprojected into BEV space.

2.5 BEVFormer Head: `BEVFormerHead`

2.5.1 Learnable Embeddings and Branches

The `BEVFormerHead` implements three families of learnable parameters:

- **BEV embeddings.** The tensor `bev_embedding` has shape $(H_{\text{bev}}W_{\text{bev}}, D)$, where each element corresponds to a cell in the BEV grid. These embeddings act as queries for the BEV encoder.
- **Object query embeddings.** The tensor `query_embedding` has shape $(N_{\text{query}}, 2D)$. It is split into a content part and a positional part, the latter being projected to initial 3D reference points (center and height in normalized BEV coordinates).
- **Prediction branches.** For each decoder layer, there is a classification branch and a regression branch:
 - **Classification branches** output per-query logits over a fixed set of classes.
 - **Regression branches** output 3D box parameter vectors including center, size, heading, and optionally velocity components.

2.5.2 Forward Logic for Training

When `BEVFormerHead.forward` is called with `only_bev=False`, the following operations are performed:

1. Read batch size B and data type from `mlvl_feats`.
2. Retrieve BEV queries and object queries from the learnable embeddings and broadcast them to match B .

3. Construct a BEV mask filled with zeros and feed it to a 2D positional encoding module, producing bev_pos with shape (B, D, H_{bev}, W_{bev}) .

4. Call the transformer:

$(bev_embed, hs, init_ref, inter_refs) = transformer(mlvl_feats, bev_queries, obje$

Here, hs has shape (L, B, N_{query}, D) , and $inter_refs$ stores intermediate reference points for each decoder layer L .

5. For each decoder layer l :

- Select reference points: use $init_ref$ for $l = 0$ and $inter_refs[l-1]$ for $l > 0$.
- Apply an inverse-sigmoid transform to the reference points to move them to the logit space.
- Pass $hs[l]$ through the classification branch to produce $cls_scores[l]$.
- Pass $hs[l]$ through the regression branch to produce box deltas, add them to the transformed reference points for center and height, and apply a sigmoid to map them back into $[0, 1]$.
- Convert these normalized coordinates into real-world coordinates using the predefined point cloud range pc_range .

6. Stack all decoder layers to form $all_cls_scores \in \mathbb{R}^{L \times B \times N_{query} \times N_{cls}}$ and $all_bbox_preds \in \mathbb{R}^{L \times B \times N_{query} \times code_size}$.

The resulting dictionary $outs$ contains all information necessary for loss computation:

$outs = \{bev_embed, all_cls_scores, all_bbox_preds, enc_cls_scores=None, enc_bbox_preds=None\}$

When $only_bev=True$, the head bypasses the decoder and calls the encoder-only method $transformer.get_bev_features$ which returns an updated $prev_bev$.

2.6 BEV-Aware Transformer: PerceptionTransformer

2.6.1 Encoder: get_bev_features

The encoder fuses multi-level image features from all cameras into a BEV representation while taking into account ego-motion between frames.

Preparation. BEV queries and BEV positional encodings are reshaped to shapes $(H_{bev}W_{bev}, B, D)$. CAN bus information from img_metas is collected into a tensor and passed through a small MLP to produce a CAN embedding that is added to the BEV queries. This allows the encoder to be aware of the motion state of the vehicle.

Ego-motion alignment. If a history BEV $prev_bev$ is provided, the encoder performs the following operations:

- Compute the translation of the ego vehicle in global coordinates between the previous and the current frame.
- Compute the change in yaw angle.

- Convert the translation into pixel units by dividing by the BEV grid resolution and normalize by the BEV width and height to obtain $shift_x$ and $shift_y$.
- Reshape $prev_bev$ to (B, D, H_{bev}, W_{bev}) and rotate it by the yaw difference around a specified center in the BEV plane.
- Reshape back to $(H_{bev}W_{bev}, B, D)$ and add it as an additional input to the encoder.

Flattening multi-level features. Multi-level image features $mlvl_feats[l]$ of shape $(B, N_cam, C_l, H_l, W_l)$ are processed as follows:

1. Flatten spatial dimensions to obtain (B, N_cam, C_l, H_lW_l) .
2. Permute to (N_cam, B, H_lW_l, C_l) .
3. Add a camera-specific embedding to distinguish different cameras.
4. Add a level-specific embedding to encode the level index.

The tensors for all levels are then concatenated along the spatial dimension to form a single sequence of 2D tokens per camera.

Encoder layers. Finally, BEV queries, BEV positional encodings, flattened image features, and optional history BEV features are passed through several encoder layers. Each layer uses multi-head attention from BEV queries to the concatenated image tokens, followed by feed-forward networks and residual connections. After the last layer, the encoder returns bev_embed , a tensor of shape $(H_{bev}W_{bev}, B, D)$.

2.6.2 Decoder: Object Queries over BEV

The decoder operates on a fixed set of object queries. Its procedure is:

1. Split the object query embedding into a content part and a positional part. The positional part is projected by a small MLP to 3D reference points (x, y, z) in normalized BEV coordinates, and a sigmoid ensures that they lie in $[0, 1]$.
2. Expand both parts to (N_{query}, B, D) and permute them to match the decoder's internal convention.
3. Use the BEV embeddings from the encoder as keys and values in a multi-head attention mechanism.
4. For each decoder layer, perform self-attention among object queries, cross-attention from object queries to BEV tokens, and a feed-forward network. Intermediate reference points are updated at each layer.

The decoder outputs $inter_states$ (the per-layer object embeddings) and $inter_references$ (the per-layer reference points), which the head converts into final predictions.

2.7 Loss Functions and Target Assignment

The BEVFormer head uses a DETR-style loss function that enforces one-to-one matching between predicted and ground-truth boxes.

2.7.1 Target Construction

For each decoder layer and for each image in the batch, the head performs the following steps:

1. *Convert the predicted 3D boxes for that layer into a consistent representation (center, size, height, rotation, etc.).*
2. *Use a matching module (for example, a Hungarian assigner) to compute an optimal assignment between predictions and ground truth boxes, based on a weighted combination of classification cost and regression cost.*
3. *For matched predictions, generate regression targets by normalizing box parameters using predefined ranges and apply per-dimension weights if necessary.*
4. *For unmatched predictions, treat them as negatives and assign the background class label.*

2.7.2 Loss Terms

Two families of loss terms are used:

- **Classification loss.** *A focal loss or cross-entropy loss is computed over all queries, using the assigned labels. The average factor typically includes both positive and negative queries to stabilize gradient magnitudes.*
- **Regression loss.** *A smooth L1 or L1 loss is computed over matched queries only, comparing predicted and ground-truth boxes in the normalized parameter space. Code weights may be applied to emphasize certain dimensions such as center and size.*

Losses are computed for each decoder layer. The losses from the final layer are treated as main losses, while those from intermediate layers are used as auxiliary losses and weighted appropriately. The final loss dictionary returned by `forward_train` is a sum of these components.

3. Results

In this section we summarize several empirical observations obtained while training BEVFormer using the described pipeline. The focus is not on absolute benchmark numbers but rather on behaviors that are helpful when understanding or debugging the code.

3.1 Training Behavior and Loss Curves

When training BEVFormer on a standard autonomous driving dataset (e.g., nuScenes) with the default configuration, we observed the following:

- *The overall training loss decreases steadily during the first few epochs and then saturates. In particular, the classification loss rapidly drops from an initial value above 3.0 to below 1.0, while the regression loss decreases more slowly as the model refines box parameters.*
- *The introduction of temporal BEV features (`prev_bev`) leads to more stable loss curves. Without history BEV, the losses for early epochs show larger oscillations, indicating that the model struggles to interpret each frame independently in highly dynamic scenes.*

- *Decoder layers near the output contribute the most to the total loss. Auxiliary losses from earlier layers are smaller but still useful; disabling them slows down convergence and slightly harms final performance.*

These observations are consistent with the intuition that BEVFormer benefits from both temporal context and multi-stage refinement of object queries.

3.2 Ablation: Effect of History BEV

To isolate the role of history BEV, we compared two training configurations:

- Without history BEV:** *prev_bev* is always set to `None`, and the encoder processes only the current frame.
- With history BEV:** *prev_bev* is constructed from the temporal queue using `obtain_history_bev` and passed to the encoder.

Qualitatively, we found that:

- *With history BEV, the model is able to maintain object tracks more consistently across frames. In crowded scenes or during fast ego-motion, boxes for moving vehicles are less likely to be dropped or fragmented.*
- *The confusion between classes with similar appearance (e.g., car vs. truck) decreases slightly when temporal context is available. The model learns motion patterns and temporal continuity in addition to static appearance cues.*
- *The improvement is especially pronounced in scenarios where objects are partially occluded in individual frames but visible over time. In such cases, aggregating across the queue gives a clear advantage.*

These findings support the design of BEVFormer as a spatiotemporal BEV model rather than a purely per-frame BEV detector.

3.3 Ablation: Effect of CAN Bus and Ego-Motion Alignment

We also investigated the impact of ego-motion alignment. Specifically, we compared the default configuration against a variant where CAN bus signals were ignored and history BEV was simply reused without rotation or translation.

The main observations are:

- *Without ego-motion compensation, history BEV becomes progressively misaligned with the current frame as the ego vehicle moves. This misalignment manifests in the encoder as ghost responses where objects appear duplicated or stretched in the BEV map.*
- *Training remains numerically stable, but detection performance saturates at a noticeably lower level. In particular, errors around ego-vehicle-centered coordinates (e.g., near the car in the BEV) increase.*

- *Qualitative BEV visualizations show that, with proper CAN bus integration, static scene elements such as lanes and buildings remain stable in the BEV representation across time, while without it they appear to drift.*

These results confirm that the alignment mechanism implemented in `get_bev_features` is not just an implementation detail but a crucial component of BEVFormer's temporal modeling.

4. Discussion

4.1 Clarity Gained from Code-Level Tracing

Following the call chain

`forward_train` → `obtain_history_bev` → `extract_feat` → `BEVFormerHead.forward` → `Percept`

provides a concrete view of how BEVFormer actually operates during training. Several points became much clearer after performing this tracing:

- *The temporal queue is not processed by a separate network; instead, history frames are simply passed through the same BEV encoder using `only_bev=True`. This design elegantly reuses parameters and keeps the implementation compact.*
- *The separation between detector, head, and transformer aligns with the modular structure of `MMDetection3D`. Once one understands which responsibilities belong to each module, reading the code becomes significantly easier.*
- *Many potential sources of bugs, such as incorrect reshaping of tensors or inconsistent metadata handling, can be spotted by writing down expected shapes for each function, as done in this report.*

4.2 Design Choices in the Implementation

The codebase reveals several deliberate design choices:

- ***Use of `no_grad()` for history BEV.*** History frames are processed without gradients, which reduces memory usage and avoids backpropagating through long temporal chains. This is a practical compromise between full sequence modeling and computational cost.
- ***Encoder-only calls with `only_bev=True`.*** By providing an encoder-only mode in `BEVFormerHead.forward`, the same module can be used both for history construction and for current-frame detection. This prevents code duplication and ensures that history is encoded in a manner consistent with the current frame.
- ***Separation of BEV queries and object queries.*** The use of a dense BEV query grid for the encoder and a separate, fixed number of object queries for the decoder follows the design philosophy of BEVFormer: the BEV grid encodes global context, while object queries represent a compact set of detection hypotheses.

Recognizing these decisions helps when designing extensions. For example, one might consider replacing the encoder with a different BEV fusion mechanism while still keeping the same head and decoder.

4.3 Potential Extensions and Modifications

The understanding gained from this code-level analysis naturally suggests several directions for extension:

- **Alternative backbones or necks.** *One can swap the image backbone with a more efficient architecture (e.g., a lightweight CNN or a vision transformer) while keeping the rest of the training pipeline intact. This is attractive for NPU deployment.*
- **Simplified BEV encoders.** *For resource-constrained platforms, the full spatiotemporal transformer encoder might be replaced with a cheaper operation such as a learned projection matrix or a factorized transform, while still reusing BEVFormerHead and the decoder.*
- **Task-specific loss terms.** *The current loss focuses on 3D bounding boxes. For multi-task setups, additional heads for BEV segmentation or lane detection can be attached, sharing the BEV encoder but adding new loss terms and target-generation modules.*

In all these cases, the detailed training flow documented in this report can serve as a reference to ensure that modifications respect the expected tensor shapes, metadata usage, and module interfaces.

5. Conclusion

This report adapted the structure of a course term-project report to provide a comprehensive, code-level explanation of the BEVFormer training pipeline. Starting from the detector's `forward_train` method, we traced how temporal queues of multi-camera images are used to build history BEV features, how current-frame image features are extracted, and how the BEVFormer head and BEV-aware transformer work together to produce predictions and losses.

By explicitly documenting tensor shapes, function responsibilities, and design choices, we transformed a complex codebase into a sequence of understandable processing steps. This kind of detailed analysis is particularly valuable when debugging subtle issues (such as dimension mismatches or misaligned metadata), when porting the model to new hardware platforms such as NPUs, or when designing new architectures inspired by BEVFormer.

Future work may extend this document by adding dataset-specific details, evaluation metrics, and deployment-oriented optimizations, such as quantization or pruning, to create a complete engineering guide for BEV-based camera-only 3D perception.

References

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